SIMATS Online Programming Lab

DS PROGRAMMING

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QUESTIONS 10 / 10 completed

Sparse Matrices

SPARSE MATRICES

Represent two sparse matrices (mostly zeros) using arrays of triples [row, col, value]. Read dimensions and non-zero terms for two matrices. Add them, store result in sparse form (sorted
by row then col), and display dense matrix form. Max 10x10; ignore zeros.

Sample Input (Matrix 1):
3 3 3
0 1 5
1 0 8
2 2 2
Sample Input (Matrix 2):
3 3 2
0 1 3
2 2 7
Sample Output (Sum Dense):
0 8 0
8 0 0

PROGRAM EDITOR

C

Run Save

```
int main(){
  int a[10],k,i,j,temp;
  printf("enter 10 no:\n");
  for(i=0;i<10;i++){
    scanf("%d",&a[i]);
  }
  printf("enter k:");
  scanf("%d",&k);
  for(i=0;i<k;i++){
    temp=a[0];
    for(j=0;j<9;j++)
      a[j]=a[j+1];
    a[9]=temp;
  }
  printf("\nrotated array:\n");
  for(i=0;i<10;i++){
    printf(" %d",a[i]);
  }
  return 0;
}
```

OUTPUT

```
enter 10 no:
enter k:rotated array:
 4 5 6 7 8 9 10 1 2 3
```

INPUT

```
1 2 3 4 5 6 7 8 9 10
3
```